

MUHAMMAD AKHTAR

Senior Software Engineer — Full-Stack (PHP/Laravel | MERN | Python) | AI/LLM & Agentic Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience across three major stacks — PHP/Laravel, the MERN/MEAN JavaScript ecosystem, and Python (FastAPI/Django/Flask) — delivering full lifecycle software from architecture to production. Proven track record building REST/SOAP API integrations, data synchronization platforms, e-commerce and CRM systems, mobile applications, and AI/LLM-powered features that automate manual work and reduce operational cost. Experienced in system design, CI/CD, cloud deployment (AWS), and mentoring engineering teams. Hands-on with AI-assisted development workflows using Claude Code and Cursor to accelerate delivery.

CORE TECHNICAL SKILLS

Backend / Languages:	PHP, Laravel, CodeIgniter, Lumen, Python, FastAPI, Django, Flask, Node.js
Frontend / Mobile:	React, Next.js, React Native (Expo), Redux Toolkit, AngularJS, jQuery, Bootstrap, HTML/CSS/JS
Databases:	PostgreSQL, MySQL, MongoDB, Redis, BigQuery
Cloud / DevOps:	AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Git, Celery, Unit Testing (PyTest)
AI / LLM:	OpenAI API, Anthropic Claude, LangChain, Agentic AI, Prompt Engineering, LLM-based Entity Resolution
Integrations:	REST APIs, SOAP/XML, OAuth 2.0, Webhooks, WebSockets (real-time notifications), Elasticsearch, Algolia, n8n Workflow Automation
AI-Assisted Dev:	Claude Code, Cursor, GitHub Copilot
Other:	System Design, ERPNext, Supabase, Payment Gateway Integration (PayPal, DIBS), Mentoring & Technical Leadership

PROFESSIONAL EXPERIENCE

Vaival Technologies — Lahore, Pakistan	Oct 2024 – Present
Senior Software Engineer	
Wash Cycle Laundry (Laundry ERP): Developed and extended a Frappe-based laundry ERP including an RFID scanning module; built and deployed a FastAPI data synchronization platform on AWS EC2 harmonizing data from multiple third-party systems into MongoDB and BigQuery, with real-time webhook integrations, GitHub Actions CI/CD, and unit testing via Pytest. WCL Driver Pro (Mobile App): Built a React Native/Expo mobile application for delivery drivers with barcode scanning, offline image uploads, and real-time tracking; integrated Redux Toolkit for state management against a Frappe backend with secure token-based authentication. n8n Workflow Automation: Orchestrated an employee data synchronization system integrating ERPNext, Everee, Supabase, and Zoho Shift using real-time webhook triggers and scheduled cron jobs.	

- **Eviction Compliance:** Researched and integrated Yardi's SOAP/XML APIs via a reusable Python library wrapping legacy SOAP services into REST/JSON interfaces; implemented Celery + Redis scheduled data synchronization.
- **RaiseIQ (Data Enrichment Platform):** Architected a multi-campaign political donor data enrichment platform using FastAPI, PostgreSQL, Redis, and Next.js; built an enrichment engine integrating RocketReach and Apollo APIs with Redis-backed fallback queues and a Chrome Extension for data collection; implemented entity resolution using PostgreSQL pg_trgm, fuzzy matching, cleanco, and libpostal; integrated Anthropic Claude to resolve ambiguous entity matches via a confidence-based workflow, reducing unnecessary LLM API calls.

Vaival Technologies — Lahore, Pakistan

Apr 2023 – Sep 2024

Software Engineer

- **Sendpad (AI-Powered Email Marketing):** Integrated OpenAI LLMs into a Laravel-based platform to generate personalized email subjects/content; designed prompt construction logic and processed OpenAI responses into ready-to-use email drafts; built template editing and customization features.
- **Cartzy eCommerce:** Maintained and enhanced Laravel REST APIs; integrated Elasticsearch for optimized product search; implemented cron-based Walmart product imports and an Ideogram logo-generation microservice.
- **Abandoned Cart Recovery:** Built a Laravel-based Shopify application to identify abandoned carts and trigger recovery emails via Amazon SES; integrated Mixpanel, Hotjar, and SpamAssassin for engagement tracking and spam analysis.

Bitsoft Solutions Private Limited — Punjab, Pakistan

Jan 2020 – Mar 2023

Software Engineer

- **Heyfocus:** Performed a security overhaul in custom PHP, replacing raw MySQL queries with PDO parameterized queries to strengthen protection against SQL injection.
- **iMakpools:** Integrated Zoho CRM using OAuth 2.0 in CodeIgniter for secure bidirectional data synchronization; integrated DIBS and PayPal payment gateways for online transactions.
- **Mindpocket:** Integrated the BigBlueButton API in CodeIgniter to enable virtual classroom functionality; resolved application issues focused on the Quiz Module; developed frontend components with HTML, Bootstrap, and jQuery.
- Conducted R&D on Zoho REST APIs, DIBS, and PayPal payment gateway integrations, preparing implementation documentation for internal use.

Gray Matrix — Remote

Apr 2020 – Sep 2020

Full Stack Engineer

- **EWS GrowHow:** Implemented multilingual support across 15 languages in Laravel, expanding platform accessibility and international reach.
- **HubStaff Replica:** Developed a Lumen microservice to receive and process bulk screenshots and activity data, storing images and metrics via Amazon S3.
- **Daily Insights:** Built Node.js real-time notification systems using WebSockets to improve employee engagement, with an AngularJS frontend; developed a birthday wishes module and integrated the Algolia Search API.

Fiverr — Remote

Oct 2019 – Dec 2019

Web Developer

- Delivered freelance web development projects for international clients across PHP and JavaScript stacks.

EDUCATION

Bachelor of Science, Computational Science — University of Southern Punjab, Multan

2015 – 2019